

Research Assignment 2: Business Analytics Terminology

(01.) Foundational Concepts of Analytics

① Business Analytics - Is the use of data and modern technology to understand how business operates and decides what actions will drive better results.

For example business analytics does not answers question like what happened but what should be organisation do with the data it has.

Mastersindatascience (2025)

② Data Analytics - Is the process of using raw data in order to see trends, and support decision making. (eg) Data analytics answers questions like what happened and what trends are in the data.

investopedia.com (2025)

③ Business Intelligence - Study ^{& Present} past data to recognise and propel current business operation, which means BI only deals with present business operation unlike BA which deals with future business operation. coursera.org (2025)

④ Data-Driven Decision-Making - Is a business approach that emphasise evidence from data and review instead of gut-feeling to inform the business, hence it uses reviewing methods such as descriptive, Diagnostic, Predictive analysis. (ibm.com, 2025)

⑤ Big Data - Is the different volumes of structured and unstructured information generated daily by humans and machines that total to huge content daily. Big Data is normally characterised by the different amount of data, the size of the data and as well as the speed at which information is received (oracle.com, 2024)

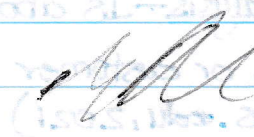
⑥ Descriptive Analysis - Is about looking at the past and present without (eg) customer sales or customer behaviour, without understanding the reason behind. (online.hbs.edu, 2021)

⑦ Diagnostic Analysis - Is the process of identifying causal factors and correlation within past data to determine the reason behind specific outcomes. (eg) Understanding why your laptop freezes when using your wireless mouse only to find out that the batteries for the mouse are finished and need to be replaced. (ibm.com, 2018)

⑧ Predictive Analysis - Is about being able to identify trends and make informed decision about what might happen in the future. (eg) Predicting hot beverage sales during winter. (ibm.com, 2018)

⑨ Prescriptive Analysis - Is not only about predicting what might happen, but also about advising what to do next (eg) How to allocate resources for the next coming projects. (ibm.com, 2018)

⑩ Data Literacy - Is the skill of understanding and using data in everyday work to guide in making better decision. The skill data literacy is being able to read data, work with data, examine the data and engage with data through (eg) story-telling. (ibm.com, 2018)

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② Data, Measurement and Reporting

① Structured Data - Is information arranged into predefined pattern normally it would be store in a table format such as database. Structured Data works well with machine but has limited flexibility and usability. (e.g) A SQL database containing customers names, ID etc, essentially customers details. (IBM.COM, 2026)

② Unstructured Data - Is information that does not have a predefined pattern, Structured data contain both written and unwritten data of which your written data would be emails, social media post and unwritten would be your images, videos. It best managed with Data links which handles large amount of data in any format. It is also stored the way it is, and is easy and cheap to store, but require expertise to prepare and examine it. (IBM.COM, 2026)

③ Data Warehouse - Is a centralized database where business merge and store important information concerning their customers, the sale of their data so that it can be used when they analyse, do their reporting and forecast their business. (e.g) of Data warehouse, could be Shoprite data warehouse where all the shoprite store of information is stored including their sales, customer loyalty program details and data of what their customer like to buy in order to understand their buying habits in order to manage their stock. Data warehouse used to store data from different sources where as Data lake is used for machine learning (Coursera.org, 2026).

④ Data Mining - Is the processing of pulling ^{useful} ~~useless~~ information that is of important from large dataset, using machine learning and other tools to reveal patterns, anomalies and insight for decision-making. Its purpose is to turn everyday information into insights that influence smarter business decisions. Data mining pullout useful information, machine learning is one of the tools used in Data mining

Data mining is used to notice unusual purchases from unfamiliar locations. Data mining ^{helps} prevent unauthorised access (eg) service history reveals more equipment failure in cold weather, suggesting that better temperature control could cut downtime (Sap.com, 2026)

(15) Data Visualisation - Converts information into pictures that makes it easy to see what is happening, spot any anomalies and explain results to others. Data visualisation is important because by turning data into visuals organizations can easily see what matters most and act on it. Mostly it helps business to recognize what affects customers choices, identify products that needs attention / improved and decide when and where to launch, and estimate future sales. (eg) Basic visuals such as bar and pie charts are easy to use, but when telling deeper data stories, tools like heat maps reveal better insights. (techtarget.com, 2026)

(16) Dashboard - Is a centralized display that brings together different types of visual data such as KPI, and business metrics into a single, easy-to-read view. Its purpose is to assist stakeholders quickly understand critical information and make informed decisions. Unlike report where they tend to focus on single event, dashboard shows high-end level view of broad amount of data in such a way they answer questions like "how was last month sales" and we have different types of Dashboard e.g Business, Executive, KPI and Project Dashboard etc all of them serving different information according to its purpose. (tableau.com, 2026)

(17) Metrics - Is a calculated value that measures business performance. It applies a consistence formula to data overtime - such as revenue, conversion rate, using a defined time frame and rules for slicing the data. Metrics transform raw number into identifier that can be compared across periods, products, regions.

⑦ Metrics - A strong metric is more than just a number it is a clearly defined agreement between business users and data professionals. When compared Metrics are the numbers you track while KPIs are the goals. Metrics purpose is to establish standard ensures organizational alignment, while accelerating decision-making, making prediction clearer, Facilitating onboarding, and allowing data teams to focus on refinement rather than enhancement (eg) Common business metrics are Revenue, conversion rate and customer churn rate etc. (IBM.COM, 2021)

⑧ Key Performance Indicator - Are Important, calculated measures that track progress towards a specific outcome. KPI helps organisations assess impact, allocates resources wisely and prioritize improvement where they matters most. They serve as foundation for both strategic and operational decisions. KPIs help you plan ahead with leading indicators and review results with lagging ones. (eg) Revenue growth Rate - Which measures how quickly sales are increasing over time. Net Profit Margin, Return On Investment and operating cash flow. KPIs differs by industry (eg) and function (eg) Finance team focuses on profitability and while HR track retention and engagement.

⑨ Benchmarking - Benchmark is the predefined standard. Is the target the business has set outlined for themselves. Benchmarking shows what's possible by comparing your results to your the compare organisations past performance, competitors or industry leaders (eg) we have competitive, internal and functional Benchmarking. Benchmarking says how well are we doing compared to the others while segmentation looks for ^{who are our} different groups of customers or data points (asana.com, 2025)

⑩ Segmentation - In the term of product management and operations - Is the process of dividing markets into different groups of buyers who have various needs, characteristic or behaviours. These groups are generally homogeneous yet distinct from the others. The purpose of segmentation is to uncover opportunities for product differentiation and to tailor

②0 Segmentation - strategies that address the unique requirement of each group. We have different types of segmentation such as psychographic, behavioural, demographic and geographic. Segmentation is very important in product management. E.g. Apple will follow with psychographic segmentation to target customer end-users that value innovation, design and quality. (Launch.notes.com, 2026)

③ Customer Sales and KPIs - Refer the plans ^{& Process} made employed by the organization to attract New customers, thereby ensuring a sustainable foundation for long-term business expansion. The benefits of

③ Customer and Sales KPIs

③1 Conversion Rate - ^{Customer Acquisition Cost} Is all about the plans and process made by the organization to attract new customer, and ensure the foundation of long-term business expansion. Conversion rate focuses on the percentage of people who convert, while Customer acquisition cost checks the total expense per customer (e.g) of conversion rate is the customer experience, e-commerce and marketing and sales. (IBM.COM, 2026)

③2 Customer Acquisition Cost -

Customer & Sales KPIs

(21) Conversion Rate - Is the ratio of users who completes the desired action of clicking an AD, installing an app, or making purchase. It reflects how effectively campaigns or end user journey turns into audiences into active customers. Conversion rate ~~are~~ provide a powerful way to compare the performance of different advertising channels. Conversion rate essentially measure how many people say "YES" while customer acquisition cost measure how much money it takes to get a "YES". The goal of conversion rate is to enhance user experience by streamlining the website and removing points of friction. (adjust.com, 2026)

(22) Customer Acquisition cost → Represents the total investment required to acquire new customers. It includes expenses tied up to marketing and sales efforts - such as advertising spend, salaries for staff, and software tools. CAC is often time evaluated alongside customer lifetime value to balance cost against long-time value/revenue. CAC & CLV combined provide insights into profitability, guide budget allocation, & help predict future growth patterns. Formula for calculating (CAC) equals Total marketing cost for acquired customer divide by Total customer acquired. The importance of CAC is to make sure you don't spend more to get customers than they'll ever bring (eg. Spending R300 for someone worth R200 that means you are losing money. (adjust.com, 2026))

(23) Customer Lifetime Value - The total profit a business can expect from a customer throughout their duration of their relationship. It is important for assessing customer experience and long-term value. CLV shows how loyal customers are and how much churn you face, and helps organisation decide where to invest. NPS vs CLV - measures total value of a customer while NPS show likely customers are to recommend your product/organisation. hence CLV formula is customer value multiple by organisation's average customers lifespan

(SaaS) Software as a service, subscription

②3 CLV — For example a UX designer subscribes to a service with multiple pricing plans. On average the customer would spend R380 per month. A typical subscription last four years on a monthly payment model, which means after calculation the value is R18240.

②4 Churn Rate — Is the ratio of customers who discontinue a service or employment who leaves within a given period. Churn rate serves as a measure of retention and overall business health. A higher churn rate hinder the growth and make scaling more difficult. The pros of churn rate is; it will show good the business really is but the negative side it does not show that the people leaving are fresh-signups or loyal, long-term customers. Churn rate will answer question such as how many existing customer did we loose while customer retention rate ask how many existing customer stayed. (Investopedia.com, 2026)

②5 Customer Retention Rate — Measure the proportion of customer a business successfully retains over a given period, where higher rate means you are keeping them happy and engaged. Retention Rate matter because keeping customer is cheaper than getting new ones, (eg) Higher rate means happy customers, low rates means there are problem that needs fixing. Customer Retention Rate are calculated by subtracting the number of new customer acquired during a period from the total customers at the end of that period, divide the result by the number of customers at the beginning of the period, then multiply by 100 to express the ratio. (Launchnotes.com, 2026)

②6 Net Promoter Score (NPS) — Is a customer experience metrics that helps organisation weigh loyalty and satisfaction with their products and services. A strong NPS often reflects a healthy business — one that customer values insights, deliver offerings people want and provide reliable customer supports. Organisations view NPS as a strong indicator of customer satisfaction. It highlight not only loyal customers

Q6) NPS - but also those who actively contribute to growth through positive word-of-mouth advocacy. Calculating NPS is simple organisations ask customers to rate how likely they are to recommend a company's product or service. Responses are typically given on a numerical scale, most often from 1 to 10. Ways of gathering NPS responses include; Email survey, on-site pop-up, SMS & text message as well as post-call survey, while NPS uses these various methods to ~~get~~ weigh customer experience (CSAT) Customer Effort score compute the ratio of responded who claim to be satisfied in a survey (ibm.com, 2026)

27. Average Order Value (AOV) - Is the typical amount spent per transaction within a specific period. Retailers use it to assess basket size, the impact of promotions, and overall customer spending patterns. A declining AOV often signals tighter consumer budget, trade-down behaviour or increased reliance on smaller purchases and discounts.

AOV becomes security reliant / relevant when used as a signal to show case anomaly, Fraud triage, or automated decision-making. Misinterpreting this metrics can cause business to overlook trends such as coupon-misuse, reseller-activity, bot-driven checkouts or synthetic customer behaviour. These blind-spots weakens frauds control and distort investigations that rely on transaction trends. (eg) would be an ecommerce team monitors AOV after introducing a free delivery threshold to determine whether customers add extra items to their cart instead of abandoning checkouts (nihil.org)

Q8. Sales Funnel - Visually maps the customer data / journey, illustrating the stages prospect move through on their way to becoming paying customers. The funnel shape reflects how the number of prospects narrows at each stage as they progress towards conversion. Challenges of Sales funnel would be; low conversion rates, high drop offs and weak leads etc and the benefits are stronger customer ^{ten} retention, Actionable Insight, Sharper targeting (etc). Best practise for building sales Funnel is Define Buyer Personas clearly, Establish metrics and KPIs, Nurture leads in the

②⑧ - In the interest stage and optimizing Top-of-funnel Content (etc)
(ibm.com, 2026)

(04) Marketing & Campaign Analytics

②⑨ Marketing campaign - Are set of planned steps business take to promote products and reach goals within a given timeline.

Types of marketing campaigns are:- Awareness focused campaign which help introduce a brand to a wider audience market, Acquisition campaign - encourage potential customers take a specific action such as signing-up new contracts. etc, and we also have ways to make campaign cut through the noise is through ① customer tailored experience - understand customer changing needs and preference as they happen and delivering content, products, or offers that match their interests. and challenges are competition in a saturated markets. (Salesforce.com, 2026)

③① Post-campaign Analysis - Is the process of reviewing completed public relation campaign to evaluate its effectiveness. It involves comparing the campaign's actual results with the objectives set at the beginning and analysing the data and feedback to understand what worked well and what could be improved for future campaigns. Post-campaign ^{analysis} are important because they help determine whether public relations campaign achieved its intended objectives. IN PR success is not only determined or measured by well designed press release or active social media content but by the actual outcomes of the campaign. It provide a structured way to measure performance and understand the overall impact of the campaign. Key thing to remember about post-campaign analysis involves reviewing of completed PR campaign to determine whether achieved its objectives. (Fivable.com, 2026)

③① Incremental Revenue - Is the additional income a business earns from increased sales. It measures the extra revenue generated by a specific product, investment or marketing campaign by comparing sales before and after increase. Incremental Revenue is important on e-commerce business because it measures how effective additional investments in marketing or growth initiative are, help business identify which business campaign generate the most profit and use those insight to improve and enhance future business marketing strategies. Improving ROI involves increasing the return on investment while keeping the cost low. IR measures increased extra sales caused by a project while ROI requires cost to calculate; cost from ^{the} denominator (lifefinsight.io, 2025)

③② Return on Investment (ROI) - Is a financial measure used to determine how profitable an investment is by comparing the returns earned with the cost of the investment. It helps business assess whether an investment, such as marketing campaign is generating worthwhile returns. ROI is calculated by dividing the net profit earned from an investment by total cost of the investment. When interpreting ROI focus on whether investment generated a profit or loss. A positive ROI indicate a gain and a negative ROI indicates a loss. (e.g.) ROI is measured by ratio while IR uses the actual currency (e.g.) rands/dollars. (salesforce.com, 2025)

③③ Return on Ad Spend (ROAS) - Measures how much revenue is generated for every amount spend on advertising. It is calculated by dividing revenue generated from advertising by the total ^{advertising} cost. ROAS is important because measures advertising effectiveness as it shows how well advertising spending converts into sales or revenue. It helps business evaluate campaign performance, allocate budgets effectively and optimise their business advertising strategies for better results. ROAS evaluate advertising performance while CTR - Tells the ratio of people who click through the ad. (mailchimp.com, 2026)

34 Click Through Rate - Measure how effectively an online advertisement attracts attention by showing percentage of people who click on the ad after seeing it. CTR when compared to CR CTR shows that an advertisement attracts people, but without indicating how many click results in sales, while Conversion Rate provide a clear measurement by showing the ratio of clicks that lead to actual purchases. CTR is calculated by ~~calculating~~ dividing the total number of clicks by total number of ad impression and multiplying the results by 100. (investopedia.com, 2026)

35 Bounce Rate - Is the measurement ^{view} ratio of website visitors where users only one page and leave without further interaction. This means they do not click links, explore additional page or take any further action which can negatively affect conversion rate. Bounce Rate is important because high rate may indicate that visitors are not finding the content useful or website are not providing good user experience. Bounce Rate should be confused with exit rate. Exit rate are percentage of visitors who leave website from a specific page. High exist rate indicate a problem, as some pages are meant to be final stage in visitors journey. Several factors can affect a website bounce rate. Type of business you in, visitors geographic location, device used to access website. How one can improve Bounce Rate, if they are Technical issue fix them, ensure that your website is mobile friendly. (Mailchimp.com, 2026)

36 A/B Testing - Is a method used to compare two version of a webpage or app to see which performs better. Different end-user are usually shown one of two versions, and the results are analysed to determine which version achieves better conversion outcomes. Why should one do A/B Testing - To help business make informed changes on their websites, or apps by using data to measure the impact of different versions. It allows teams to test assumption understand which changes influence user behaviour, and

(36) Identify most effective approach for achieving the goals
A/B Testing examples: 1. Home page A/B Test, Pop-up To Flop-up etc
Digital marketing teams can improve testing results by involving different departments, helping them make informed decision and measure - the impact of marketing changes on business performance, which includes A/B Testing social media, A/B Testing marketing and Website A/B Testing. (Optimizely, 2026)

(37) Lead Generation - Is the process of attracting and identifying potential customers who may be interested in a business products or services. It involves nurturing these prospect throughout the buying journey with the goal of converting interested leads into paying customers. To generate high quality leads, marketers should create relevant and engaging message that connect with their target audience. Reaching potential customers through the channels they use most can increase brand awareness, strengthen trust and attracts More qualified prospects. To accumulate successful Lead Generation one must (eg) Identify target audience, create relevant messaging and Nurture relationship (business.adobe.com, 2026)

(38) Marketing Attribution - Is the process of determining which customer interactions such as clicks, emails, or social media post, contributed to a conversion. Helps business understand which marketing channels are generating results and where to allocate their budget effectively. Attribution has four different types of models single-Touchpoint marketing attribution, Multi-touchpoint rule-based attribution, algorithm driven / data-driven and econometric models Attribution vs casual measurement, MA shows relationship rather than direct cause and effect. They assign credit to marketing touch points based on their role in the customer journey but do not prove that specific touchpoint caused conversion. (windSOR.ai, 2026)